

5.3.20 Temperature sensor characteristics

Table 58. Temperature sensor characteristics

Symbol	Parameter	Min	Typ	Max	Unit
$T_L^{(1)}$	V_{SENSE} linearity with temperature (from V_{sensor} voltage)	-	-	3	°C
	V_{SENSE} linearity with temperature (from ADC counter)	-	-	3	
Avg_Slope ⁽²⁾	Average slope (from V_{sense} voltage)	-	2.14	-	mV/°C
	Average slope (from ADC counter)	-	2.14	-	
$V_{30}^{(3)}$	Voltage at 30° C (± 1 ° C)	-	0.65	-	V
$t_{start_run}^{(1)}$	Startup time in Run mode (buffer startup)	-	-	25.2	μ s
$t_{S_temp}^{(1)}$	ADC sampling time when reading the temperature	13	-	-	
$I_{sens}^{(1)}$	Sensor consumption	-	0.18	0.29	μ A
$I_{sensbuf}^{(1)}$	Sensor buffer consumption	-	3.8	6.5	

1. Specified by design - Not tested in production.
2. Evaluated by characterization - Not tested in production.
3. Measured at $V_{DDA} = 3.3\text{ V} \pm 10\text{ mV}$. The V_{30} ADC conversion result is stored in the TS_CAL1.

Table 59. Temperature sensor calibration values

Symbol	Parameter	Memory address
TS_CAL1	Temperature sensor raw data acquired value at 30 °C, $V_{DDA} = 3.3\text{ V}$	0x08FF F814 - 0x08FF F815
TS_CAL2	Temperature sensor raw data acquired value at 140 °C, $V_{DDA} = 3.3\text{ V}$	0x08FF F818 - 0x08FF F819

5.3.21 Digital-to-analog converter characteristics (DAC)

Table 60. DAC characteristics

Specified by design and not tested in production.

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Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V _{DDA}	Analog supply voltage	-		2.7	-		V
V _{REF+}	Positive reference voltage	-		2.5	-	V _{DDA}	
V _{REF-}	Negative reference voltage	-		-	V _{SSA}	-	
R _L	Resistive Load	DAC output buffer ON	connected to V _{SSA}	5	-	-	KΩ
			connected to V _{DDA}	25	-	-	
R _O	Output Impedance	DAC output buffer OFF		10.3	13.00	16	
R _{BON}	Output impedance sample and hold mode, output buffer ON	DAC output buffer ON	V _{DD} = 2.7 V	-	-	1.6	KΩ
R _{BOFF}	Output impedance sample and hold mode, output buffer OFF	DAC output buffer OFF	V _{DD} = 2.7 V	-	-	17.8	KΩ
C _L	Capacitive Load	DAC output buffer OFF		-	-	50	pF
C _{SH}		Sample and Hold mode			0.10	1	μF
V _{DAC_OUT}	Voltage on DAC_OUT output	DAC output buffer ON		0.2	-	V _{DDA} - 0.2	V
		DAC output buffer OFF		0	-	V _{REF+}	